K-Bound: When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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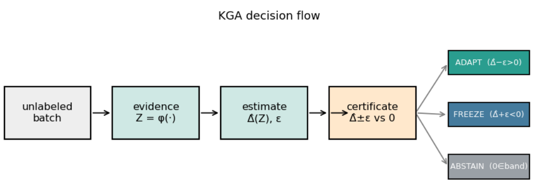


Fig. 1. **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (certified helpful), FREEZE (certified harmful), or ABSTAIN (no supported strict commitment). Under interval coverage it controls the marginal events $\mathbb{P}(\text{ADAPT}, \Delta \leq 0)$ and $\mathbb{P}(\text{FREEZE}, \Delta \geq 0)$ at level α; no conditional-error guarantee is claimed.

***Abstract*—Test-time adaptation (TTA) may help or harm on unlabeled target data, so a deployment system must choose whether to *adapt*, *freeze*, or *abstain* from committing an update. We define adaptation benefit as $\Delta = R_T(f_0) - R_T(f_a)$, so positive values favor adaptation. Over a declared calibration-drift class, a strict adapt or freeze commitment is uniformly supportable if and only if the population evidence margin satisfies $|M| > \beta$; inside the closed band, matched evidence or zero-benefit ambiguity prevents a uniformly sound strict commitment. Knowability-Guided Adaptation (KGA) addresses the separate finite-sample layer: it estimates Δ directly from label-free evidence and commits only when a calibrated interval $\widehat{\Delta} \pm \varepsilon$ excludes zero. On real benchmarks KGA does not numerically apply $|M| > \beta$, and ε is not an estimate of β. Controlled mixed regimes yield confidence-interval-supported wins over both fixed policies, whereas natural one-sided shifts mainly yield no-harm behavior and weak evidence yields abstention or diagnostic failure. K-Bound is therefore a safety and validity layer, not a universal accuracy booster.**

***Index Terms*—test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation**

I. Introduction

Test-time adaptation (TTA) updates a model on the unlabeled test stream and, under favorable distribution shift, recovers accuracy. Under unfavorable shift, the same objective silently degrades it, and with no target labels the system usually cannot tell which case it faces. The decision that matters therefore comes *before* the update: given a frozen baseline f_0 and an adapted candidate f_a, choose **adapt** when adaptation is knowably helpful, **freeze** when knowably harmful, or **abstain** when label-free evidence cannot identify the sign of the benefit. The field's default assumption is that adaptation should be made robust enough to always run; we replace it with an explicit, certified decision about whether to run it at all.

Some cases are information-theoretically unknowable. It is established that label-free quantities cannot decide adaptation without an assumption on the shift: identifying OOD accuracy is as hard as identifying the optimal predictor [11], and the optimal predict-versus-abstain rule under a shift budget has a known form [12]. We sharpen this landscape into a strict-commitment criterion: a strict adapt or freeze action is uniformly sound over the declared drift class *if and only if* $|M| > \beta$ (Theorem 2). In the interior, evidence-identical worlds can have opposite nonzero benefits (Theorem 1); on the boundary, zero-versus-strict ambiguity still rules out a uniformly sound strict action (Proposition 2). KGA (Algorithm 1), a benefit estimator plus a conformal radius wrapped around any adapter (Tent, EATA, SAR), realizes the three-way decision as a finite-sample certificate—a direct benefit estimate with a calibrated radius, not a numerical $|M|>\beta$ test—controlling marginal false-adapt and false-freeze events under interval coverage (Theorem 3).

The evidence follows the regime map in Table I. *Identifiable mixed* harmful+helpful regimes supply the **beats-both** policy wins, including a pre-registered head-to-head over protocol-matched POEM-style and AETTA-style baselines. Natural shifts primarily test safety: a reconciled result may tie the better fixed policy, while a provisional result remains outside the headline until its saved summary and canonical replay agree. K-Bound does *not* claim universal improvement: when adaptation is already safe, a valid certificate can only tie always-adapt. It provides a safety layer for TTA rather than an accuracy boost.

a) Plain-language takeaway.: If the population evidence margin dominates declared drift ($|M| > \beta$), a strict action is uniformly supportable; otherwise the maximal sound three-way rule ABSTAINS. KGA implements the distinct finite-sample interval decision. On single natural-shift datasets this mostly acts as safety or abstention. Beats-

TABLE I

Regime map: expected behavior vs. result. Each regime, an example dataset, the behavior the frontier predicts for a valid certificate, and what was measured. "No-harm" = matches the better fixed policy and beats the worse, without beating both; "beats both" = lower regret than always-adapt *and* always-freeze. The evidence tier in Table XV states whether that point verdict is CI-confirmed, reconciled, provisional, or diagnostic.

Regime	Example	Expected	Result
Helpful-dominated	Camelyon17 OOD	tie always-adapt	reconciled no-harm
Harmful-dominated	RxRx1; iWildCam and Office-Home OOF summaries	tie always-freeze, beat always-adapt	reconciled no-harm
Domain-gen. natural	PACS (4 LODO splits)	no-harm or abstain	three safety tracks; one null
Mixed + detectable	CIFAR-10-C Tent/EATA, ImageNet-C SAR	beat both fixed policies	CI-supported
Weak evidence	ImageNet-R, CIFAR-10.1	abstain / conservative	diagnostic; no promoted win
Constructed mixture	three-source OOF stream	route per condition, beat both	supported routing result, not transfer

both wins require a detectable mixed regime; a constructed pooled mixture is reported as such, not as a claim of single-dataset natural-shift dominance.

b) Contributions.:

1) **Adapt/freeze/abstain formulation.** We make the target risk difference, strict actions, fallback semantics, and target-label-free information boundary explicit.
2) **Exact strict-commitment frontier.** A strict adapt/freeze action is uniformly sound iff $|M| > \beta$; the interior and boundary cases are proved separately (Theorem 2).
3) **Coverage-based finite-sample certificate.** Under measurable interval coverage, the empirical rule controls marginal false-adapt and false-freeze events (Theorem 3).
4) **KGA deployable wrapper.** KGA wraps Tent/EATA/SAR without changing their adaptation objectives and uses shadow state, commit, rollback, and logging (Algorithm 1).
5) **Regime-based empirical evaluation.** A uniform panel separates controlled mixed wins, natural no-harm results, constructed routing, and weak-evidence diagnostics.

Table note. The uniform panel below is the sole empirical index; diagnostic variants and historical runs are not additional evidence for the headline claims.

II. Related Work

The closest literature improves test-time updates, monitors adaptation failures, estimates target performance without labels, or characterizes abstention under shift. K-Bound differs in the decision object: it certifies the sign of the frozen-versus-adapted risk difference before the candidate state is committed.

A. Test-Time Adaptation

Tent [1], SHOT [2], TTT [3], EATA [4], SAR [5], CoTTA [6], and MEMO [7] improve adaptation through entropy minimization, filtering, regularization, sharpness-aware updates, or temporal stabilization (survey: [33]). These methods primarily answer *how* to adapt. Even when the update rule is more stable, the deployment system generally still commits to adaptation rather than certifying that the candidate has positive target benefit.

B. Guarded and Monitored Adaptation

POEM [10], sequential risk monitors [8], [9], PeTTA [40], and sample-level filters such as DeYO [41] guard an adaptation stream during or after updating. These approaches are complementary to K-Bound. They typically monitor a proxy or intervene in one failure direction, whereas K-Bound poses a pre-commitment three-way decision over the benefit sign and explicitly represents an unknowable region.

C. Label-Free Performance Estimation

ATC [11], agreement-on-the-line [14], [13], AETTA [15], and related confidence-style estimators [32], [31] estimate target accuracy or error without target labels. Estimating $R_T(f)$ is not identical to certifying $\mathrm{sign}(R_T(f_0) - R_T(f_a))$: a deployment controller needs the sign of a paired risk difference and must account for uncertainty around zero. K-Bound uses this difference as the estimand and abstains when the available evidence does not identify its sign.

D. Impossibility, Abstention, and Risk Control

General label-free target-risk identification requires assumptions on the shift [11], [16], [17], [18]. Kalai and Kanade [12] characterize a related abstain-versus-predict decision under a shift budget, while selective prediction [19] abstains on individual labels. K-Bound specializes this landscape to adaptation benefit, localizes the comparison to the disagreement region, and derives an adapt/freeze/abstain certificate. Its empirical uncertainty layer connects to split conformal prediction, conformal under shift, RCPS, and Learn-then-Test [34], [36], [35], [38], [39].

E. Surviving Gap

Existing work provides stronger adapters, reactive monitors, label-free accuracy estimates, and general abstention theory. The remaining gap is a pre-commitment rule for the sign of adaptation benefit that (i) distinguishes helpful, harmful, and unsupported decisions, (ii) states the assumptions under which the decision is valid, and (iii) controls marginal false adaptation under a calibrated uncertainty model. K-Bound addresses this gap with the population frontier and its finite-sample realization, KGA.

TABLE II

DATA ACCESS BY STAGE. TARGET-TEST LABELS ARE FORBIDDEN UNTIL OFFLINE AUDIT.

Stage	Source labels	Dev labels	Target-test labels
Source training	yes	optional	no
Benefit/radius calibration	optional	yes	no
Deployment decision	no	frozen artifacts only	no
Offline evaluation	no	no	yes, after decisions

III. PROBLEM FORMULATION AND VALIDITY

A. Deployment Setting and Adaptation Benefit

Source data follow $P_S(X,Y)$ and may be labeled during training and development. Deployment data follow $P_T(X,Y)$, but only target inputs and model outputs are available when the decision is made. A frozen predictor f_0 serves as the safe fallback, and an adapter $\mathcal{A}$ proposes a temporary candidate $f_a = \mathcal{A}(f_0, X_{1:m})$ from an unlabeled target batch. For loss ℓ, the target risk and adaptation benefit are

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)], \qquad \Delta = R_T(f_0) - R_T(f_a). \tag{1}$$

Thus $\Delta > 0$ favors committing the candidate, $\Delta < 0$ favors retaining the frozen state, and $\Delta = 0$ makes the two actions risk-equivalent. Target labels reveal Δ only during offline auditing; they are not available to the deployment rule.

B. Target-Label-Free Evidence and Decision Semantics

The controller observes a label-free evidence vector $Z = \phi(X_{1:m}, f_0, f_a)$ containing quantities such as prediction entropy, confidence, class-balance statistics, frozen-versus-adapted disagreement, output-distribution drift, and update magnitudes. The deployment rule returns

$$g(Z) \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}.$$

ADAPT commits f_a; FREEZE rejects the proposal and retains f_0 (or the latest certified state in continual operation); and ABSTAIN declines to change the model, serves the frozen prediction, and may request more evidence or human review. Abstention therefore applies to the update decision, not to prediction itself.

C. What "Label-Free" Means

K-Bound is target-label-free rather than supervision-free. Source labels and a held-out development set may be used to fit the benefit estimator and calibrate its uncertainty radius. No target-test label may influence the adapter configuration, evidence map, benefit estimator, radius, action threshold, or model selection. Target-test labels are opened only after all decisions are frozen, for reporting accuracy, regret, and false-adapt events.

D. Assumptions and Validity Scope

The theory and the finite-sample certificate rely on distinct assumptions. *Risk alignment* determines whether the evidence can identify or bound the benefit sign over the declared target class. *Coverage* determines whether the empirical interval contains the true benefit at the stated rate. *Exchangeability* (or an explicit shift correction) connects calibration residuals to deployment residuals. The adapter, evidence schema, estimator, and calibration protocol must be fixed before held-out scoring. If these conditions cannot be justified, the theorem does not license a commitment; the operational fallback is freeze or abstain. Risk alignment is an assumption on the deployment class, not a property certified by fitting a benefit regressor: support and residual diagnostics may reject implausible evidence maps, but they do not verify it universally.

TABLE III

ASSUMPTIONS, THEIR ROLE, AND THE SAFE RESPONSE WHEN THEY ARE UNSUPPORTED.

Condition	Role	If unsupported
Risk-aligned Z	Benefit sign is recoverable over the declared class	abstain; do not interpret a point estimate as a certificate
Coverage of $\widehat{\Delta} \pm \varepsilon$	Controls marginal false-adapt/freeze	recalibrate, widen the radius, or freeze
Exchangeable or corrected residuals	Transfers calibration to deployment	use shift-aware calibration or abstain
Bounded loss/benefit	Supports concentration and finite-sample analysis	clip/redefine the loss or state no bound
Frozen protocol	Prevents target-test leakage and selection bias	rerun with a locked protocol

TABLE IV

CORE QUANTITIES. THE POPULATION DRIFT BUDGET β AND EMPIRICAL RADIUS ε ARE RELATED BUT NOT INTERCHANGEABLE.

Symbol	Meaning	Available at deployment?
f_0, f_a	frozen and candidate models	yes
Δ	true target adaptation benefit	no
D	disagreement region	yes
M	observable population margin	through label-free evidence
γ	realized calibration drift	no
β	declared bound on $\lvert\gamma\rvert$	supplied pre-deployment
Z	label-free evidence vector	yes
$\widehat{\Delta}$	estimated adaptation benefit	yes
ε	calibrated residual radius	fixed pre-deployment
α	marginal error level	chosen pre-deployment

E. Notation and Observability

Table IV lists the core quantities and marks which are observable at deployment.

F. Multiclass and Regression Scope

The binary 0–1 loss case gives the cleanest frontier because, on $D = \{x : f_0(x) \neq f_a(x)\}$, exactly one predictor is correct. For multiclass 0–1 loss,

$$\Delta = P_T(D)(p_a - p_0), \qquad p_j = P_T(f_j(X) = Y \mid D),$$

so the decision becomes an ordinal comparison of the two disagreement-region accuracies. For squared-loss regression, the risk difference reduces to a disagreement-local moment. The main theorem uses a split-observable decomposition $\operatorname{sign}(\Delta) = \operatorname{sign}(M+\gamma)$; the multiclass bridge is stated below and the regression derivation appears in Appendix D.

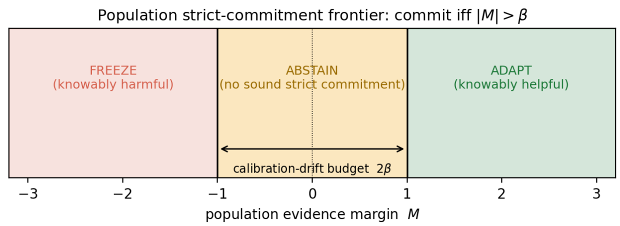


Fig. 2. **Population strict-commitment frontier.** The population rule supports adapt or freeze outside the drift band and abstains inside it. At the boundary, zero-benefit ambiguity rules out a uniformly sound strict action; opposite nonzero signs are claimed only in the interior. KGA uses the separate finite-sample interval rule in Fig. 3.

Proposition 1 (Multiclass benefit decomposition)**.** *For multiclass* 0/1 *loss,* $\Delta = \mu_T(D)(p_a - p_0)$ *with* $p_j = \mathbb{P}_T(f_j(X){=}Y \mid D)$*. If the disagreement-region accuracy gap admits a split-observable decomposition* $p_a - p_0 = M_{\mathrm{mc}} + \gamma_{\mathrm{mc}}$ *with* $|\gamma_{\mathrm{mc}}| \le \beta_{\mathrm{mc}}$*, then* $|M_{\mathrm{mc}}| > \beta_{\mathrm{mc}}$ *implies* $\operatorname{sign}\Delta = \operatorname{sign} M_{\mathrm{mc}}$*. The converse strict-commitment result additionally requires a richness assumption: the declared class must contain augmented-evidence-identical laws whose admissible* γ_{mc} *values straddle* $-M_{\mathrm{mc}}$*, together with the zero-benefit boundary law. At* $M_{\mathrm{mc}} = \beta_{\mathrm{mc}} = 0$*, the sign is identified as zero.*

In the multiclass experiments KGA estimates Δ directly through the benefit proxy $\widehat{\Delta} = h_\theta(Z)$ (not M_{mc} itself); the binary frontier is the identifiability backbone, and the multiclass results are evaluated through this benefit-estimation route.

G. Formal Setup

Source law $P_S(X, Y)$ supplies labels at train/validation; target law $P_T(X, Y)$ exposes only $P_T(X)$ at deployment. Fix loss ℓ, frozen predictor f_0, candidate f_a, and target measure μ_T on inputs. Write target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$ and the *adaptation benefit*

$$\Delta := R_T(f_0) - R_T(f_a) \qquad (\Delta > 0 \Leftrightarrow \text{adapting helps}).$$

The *disagreement region* $D := \{x : f_0(x) \neq f_a(x)\}$ is observable from (f_0, f_a).

Assumption 1 (Deployment setup)**.** Unless stated otherwise: binary label $Y \in \{0, 1\}$, 0/1 loss ℓ, and $\mu_T(D) > 0$. Fix a source-calibrated score $s : \mathcal{X} \to [0, 1]$ estimating the candidate's *pointwise correctness* (e.g. a Platt-scaled confidence that f_a is right at x), and write the target pointwise correctness $\eta_a(x) := \mathbb{P}_T\big(f_a(X){=}Y \mid X{=}x\big)$ on D, so that $\mathbb{E}_{\mu_T}[\eta_a \mid D] = \bar{a}$. Define the *observable margin*

$$M := \mathbb{E}_{\mu_T}[s(X) \mid D] - \tfrac{1}{2},$$

the *calibration drift* (unobserved at deploy)

$$\gamma := \mathbb{E}_{\mu_T}[\eta_a(X) - s(X) \mid D],$$

and label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ for any measurable ϕ using only unlabeled batches and model outputs (entropy, confidence, drift, disagreement, update norms). A *label-free rule* is any measurable map $g : \mathcal{Z} \to \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ that does not use target labels on the deployment batch.

Definition 1 (Risk alignment)**.** Evidence Z is *risk-aligned on a class* $\mathcal{P}$ if no two laws $P_T^1, P_T^2 \in \mathcal{P}$ with $\operatorname{sign}\Delta_1 \neq \operatorname{sign}\Delta_2$ satisfy $\operatorname{Law}(Z \mid P_T^1) = \operatorname{Law}(Z \mid P_T^2)$.

All frontier claims stated in terms of M additionally require M to be determined on each evidence-law fibre; equivalently, take the population evidence to be $\widetilde{Z} = (Z, M)$. Without this requirement, $|M| > \beta$ proves sign stability *given* M, not identifiability from $\operatorname{Law}(Z)$ alone. Fitting a benefit regressor does not certify risk alignment: support, dependence, and residual diagnostics may falsify a proposed evidence map but cannot verify the assumption uniformly over the declared deployment class.

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if the population frontier certifies $\Delta > 0$ ($\textsc{adapt}$); *knowably harmful* if it certifies $\Delta < 0$ ($\textsc{freeze}$); *unsupported for strict commitment* if the same augmented-evidence law is compatible with distinct signs, including zero versus a strict sign ($\textsc{abstain}$).

Remark 1 (Four quantities)*.* Do not confuse M (observable margin, deploy-time), γ (realized drift, latent), β (declared bound $|\gamma| \le \beta$ on the deployment class), and ε (finite-sample radius from held-out residuals). The radius ε operationalizes coverage; it is *not* an estimate of β.

Symbol	Meaning	At deploy?	Role
M	observable evidence margin	yes	evidence toward adapt/freeze
γ	realized calibration drift	no	can reverse the benefit sign
β	declared bound on $\lvert\gamma\rvert$	externally supplied	population frontier
ε	residual quantile radius	estimated pre-deploy	finite-sample certificate

What "label-free" means. No target/test labels at deploy; labeled source and held-out validation may calibrate $\widehat{\Delta}$ and ε. Evidence Z uses only unlabeled batches, predictions, entropy, drift, and update statistics. The method is **target-label-free**, not label-free of all supervision: labels may be used only in the source/validation calibration split, never in the deployment batch or held-out test decision.

K-Bound in one sentence. A strict adapt/freeze commitment is uniformly supportable only when the population evidence margin exceeds the declared drift budget; otherwise abstention is the maximal sound three-way action.

For a supplied drift budget $\beta \ge 0$, the *declared deployment class* is

$$\mathcal{C}_\beta := \big\{P_T : |\gamma(P_T)| \le \beta\big\}.$$

Identifiability statements below are *over* $\mathcal{C}_\beta$, not over arbitrary shifts.

IV. Theory

A. Roadmap

The theoretical spine has three layers. First, the risk difference reduces to the region where the frozen and candidate predictions disagree. Second, matched label-free evidence can support incompatible benefit signs, which establishes an abstention requirement. Third, the decomposition $\operatorname{sign}(\Delta) = \operatorname{sign}(M+\gamma)$ yields the population frontier, while a calibrated interval around $\widehat{\Delta}$ yields the finite-sample decision rule.

Lemma 1 (Disagreement-region reduction)**.** *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a{=}Y \mid D)$,*

$$\Delta = 2\,\mu_T(D)\big(\bar{a} - \tfrac{1}{2}\big),$$
$$\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \tfrac{1}{2}) = \operatorname{sign}(M+\gamma).$$

The identity $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *holds for* every *target law; it does not assume $|\gamma| \le \beta$.*

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ,

$$\begin{aligned} M+\gamma &= \Big(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2}\Big) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}, \end{aligned}$$

so $\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \frac{1}{2}) = \operatorname{sign}(M+\gamma)$. □

Theorem 1 (Interior matched-evidence impossibility)**.** *Fix $\alpha \in (0, \frac{1}{2})$ and $\beta > 0$. For any margin with $|M| < \beta$ there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ such that* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$, $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \ne 0$, *and $|M(P_T^1)| < \beta$. (For $\beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so* $\operatorname{sign}\Delta = \operatorname{sign} M$ *is determined and no such pair exists: matched-evidence ambiguity is a strictly-positive-drift phenomenon.) Consequently, any label-free rule g with $\mathbb{P}[g = \textsc{adapt}, \Delta \le 0] \le \alpha$ and $\mathbb{P}[g = \textsc{freeze}, \Delta \ge 0] \le \alpha$ must satisfy $\mathbb{P}[g = \textsc{abstain}] \ge 1 - 2\alpha$ on this shared evidence law.*

Proof. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$. Keep the input law, f_0, f_a, s, and hence $\widetilde{Z} = (Z, M)$ fixed, but set the conditional target labels so that $\bar{a}_\pm = \frac{1}{2} \pm \delta$. Then $\gamma_\pm = \pm\delta - M \in [-\beta, \beta]$ and Lemma 1 gives opposite nonzero benefit signs at the same augmented-evidence law. A label-free rule therefore has the same action probabilities in both worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world, so its abstention probability is at least $1-2\alpha$. An explicit Gaussian witness appears in Appendix C (Theorem 4). □

Proposition 2 (Boundary case and closed-band abstention)**.** *Fix $\alpha \in (0, \frac{1}{2})$. If $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world ($\gamma = -M$) and a strict-benefit world of sign M ($\gamma = 0$). In the zero-benefit world both strict actions are errors under the events used in Theorem 1, so any rule controlling both marginal errors at level α abstains with probability at least $1 - 2\alpha$. If $M = \beta = 0$, every admissible world has $\Delta = 0$ and the same conclusion holds directly. Thus abstention is the maximal sound three-way action throughout the closed band $|M| \le \beta$, although opposite nonzero signs are asserted only in the interior.*

Theorem 2 (Exact strict-commitment frontier)**.** *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \ge 0$.*

- (i) ***Reduction.*** *For every P_T,* $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *(Lemma 1).*
- (ii) ***Sufficiency.*** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then* $\operatorname{sign}\Delta = \operatorname{sign} M$.
- (iii) ***Necessity.*** *If $|M| < \beta$ (which requires $\beta > 0$), there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ with the same augmented-evidence law and opposite* nonzero *benefit signs; at $|M| = \beta > 0$ the admissible drift $\gamma = -\operatorname{sign}(M)\beta$ yields $\Delta = 0$, and in the degenerate case $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\Delta \equiv 0$. Hence no* strict *benefit direction is uniformly supported by* $\mathrm{Law}(\widetilde{Z})$ *over $\mathcal{C}_\beta$ whenever $|M| \le \beta$.*
- (iv) ***Strict-commitment iff.*** *A strict action in $\{\textsc{adapt}, \textsc{freeze}\}$ is uniformly sound over $\mathcal{C}_\beta$ if and only if $|M| > \beta$; on the closed band $|M| \le \beta$ (including $M = \beta = 0$, where $\Delta \equiv 0$) abstention is the maximal sound action.*

The population rule $M > \beta \Rightarrow \textsc{adapt}$, $M < -\beta \Rightarrow \textsc{freeze}$, $|M| \le \beta \Rightarrow \textsc{abstain}$ is the maximal sound certificate on $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 1. For (ii), if $M > \beta$ and $|\gamma| \le \beta$, then $M + \gamma > 0$, so $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma) = +1$; the negative case is symmetric. For (iii), pick $\gamma_\pm \in [-\beta, \beta]$ with $\operatorname{sign}(M+\gamma_+) \ne \operatorname{sign}(M+\gamma_-)$, both nonzero, when $|M| < \beta$; Lemma 1 yields matched Z and opposite signs. At $|M| = \beta$ the admissible $\gamma = -\operatorname{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 2. Maximality follows from Theorem 1 in the interior and Proposition 2 on the boundary. □

Region	Meaning	Action
$M > \beta$	allowed drift cannot flip the positive sign	ADAPT
$M < -\beta$	allowed drift cannot flip the negative sign	FREEZE
$\|M\| < \beta$	drift can reverse the strict sign	ABSTAIN
$\|M\| = \beta > 0$	drift can reduce benefit to zero	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

Theorem 3 (Finite-sample adapt/freeze/abstain certificate)**.** *For any real-valued target benefit Δ, suppose the benefit interval satisfies marginal coverage*

$$\mathbb{P}\big[\,|\widehat{\Delta} - \Delta| \le \varepsilon\,\big] \;\ge\; 1 - \alpha.$$

Define

$$g(Z) = \begin{cases} \textit{ADAPT} & \widehat{\Delta} - \varepsilon > 0, \\ \textit{FREEZE} & \widehat{\Delta} + \varepsilon < 0, \\ \textit{ABSTAIN} & \textit{otherwise.} \end{cases}$$

Then, marginally over the calibration randomness,

$$\mathbb{P}\big[g(Z) = \textit{ADAPT}, \Delta \le 0\big] \le \alpha,$$
$$\mathbb{P}\big[g(Z) = \textit{FREEZE}, \Delta \ge 0\big] \le \alpha.$$

Proof. On the event $\{|\widehat{\Delta} - \Delta| \le \varepsilon\}$, the condition $\widehat{\Delta} - \varepsilon > 0$ implies $\Delta > 0$, so a false adapt requires the complement of this event, which has probability at most α. The freeze case is symmetric. □

Exchangeability, or an explicit shift correction, is one route to this coverage assumption. Risk alignment is not required for the conditional-on-coverage implication.

Remark 2 (Marginal vs. conditional error). Theorem 3 bounds the *marginal, unconditional* false-adapt and false-freeze probabilities ($\mathrm{FA_u}$). It does *not* bound the conditional rate $\mathrm{FA_c} = \mathbb{P}[\Delta \le 0 \mid \text{ADAPT}]$, which we report empirically. Risk alignment (Definition 1) is required for the *frontier* identifiability claim; coverage alone certifies the *interval* rule above.

Corollary 1 (Unjustified assumptions ⇒ abstain)**.** *If interval coverage—or the calibration assumptions supporting it (exchangeability, or an explicit shift correction)—cannot be justified on the deployment batch, Theorem 3 does not apply, and committing to* ADAPT *is not protected at level α; the conservative action is* ABSTAIN *or* FREEZE*. If instead* risk alignment *(Definition 1) cannot be justified, the population-frontier interpretation and the transferability of the benefit estimator are unsupported, even though the interval implication of Theorem 3 remains valid* conditional on coverage *(cf. Theorem 1).*

B. How the Results Fit Together

disagreement-region reduction → matched-evidence ambiguity → **Thm. 1** (abstention necessity)
→ split-observable decomposition sign $\Delta = \mathrm{sign}(M + \gamma)$ → **Thm. 2** ($|M| > \beta$ frontier)
→ coverage condition → **Thm. 3** (finite-sample certificate)

C. Extensions and Deferred Results

Additional minimax, anytime-valid, multicandidate, and one-bit results are deferred to a future supplement and are not used by the short-paper claims. The saved iWildCam streaming script computes its betting increments from frozen and adapted window accuracies, which require target labels. We therefore retain that run only as a label-informed offline stress diagnostic; it is not a label-free KGA deployment result and no streaming empirical claim is made in this paper.

D. Claim Scope

The population theorem is conditional on the declared class $\mathcal{C}_\beta$, and the finite-sample guarantee is conditional on valid interval coverage. The theorem controls the marginal probabilities $\mathbb{P}(\text{ADAPT}, \Delta \le 0)$ and $\mathbb{P}(\text{FREEZE}, \Delta \ge 0)$; it does not automatically control the conditional error among committed actions. The framework does not claim that β is identifiable from unlabeled deployment data, that the empirical radius ε estimates β, or that calibration coverage transfers to arbitrary unseen domains without an assumption or correction.

V. KGA Method

A. System Overview

KGA is the paper's practical finite-sample method—a decision wrapper around a fixed adapter, not a new adaptation objective and not a numerical implementation of the population $|M| > \beta$ rule. We fit the benefit estimator and its radius *out of fold*, so the data used to fit $\widehat{\Delta}$ and the data used to calibrate its residual are disjoint for every point. On the per-cell stress grids (CIFAR-10-C and the synthetic grids), for each cell i a gradient-boosted $\widehat{\Delta}_{-i}$ is trained on the other $N-1$ cells and evaluated on cell i, and ε is an upper empirical quantile of the leave-one-out residuals $\{|\widehat{\Delta}_{-i}(Z_i) - \Delta_i|\}_{i=1}^{N}$—leave-one-condition-out cross-fitted empirical residual calibration, so no cell's residual uses an estimator that saw it. On the natural-shift protocols (Camelyon17, iWildCam, Office-Home) the split is across domains: $\widehat{\Delta}$ and ε are fit once on a dev split and the untouched held-out target-domain conditions are scored a single time. At deploy we extract Z from an unlabeled batch and return adapt, freeze, or abstain (Algorithm 1). Labels enter only through the calibration split; the deployment batch and held-out test scorer never use target labels for the adapter, ε, evidence, or rule selection. The coverage guarantee is split-dependent. The current released scorer uses the exact split-conformal rank $\varepsilon = r_{(k)}$, $k = \min\{n_{\mathrm{cal}}, \lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil\}$. Under exchangeability this supports finite-sample marginal coverage on a clean split. The archived benchmark JSONs, however, were generated by the earlier interpolated empirical quantile; they are therefore reported as empirical benchmark evidence rather than retroactively relabeled exact split-conformal runs. The per-cell grids set ε to the $(1-\alpha)$ empirical quantile of the leave-one-out residuals, so the *calibration-set* coverage is approximately nominal (realized 0.898 at nominal 0.90, α=0.10)—empirical calibration evidence, not an exact finite-sample statement; this is the standard jackknife radius, which is distribution-free only asymptotically under estimator stability; its formal finite-sample counterpart is the jackknife+ of Barber et al. (2021) at level $1 - 2\alpha$, which we note but do not implement. The decision rule's false-adapt guarantee is stated under the coverage hypothesis (§V), which the empirical coverage approximately meets. This construction

TABLE V

Label-free evidence in the promoted per-condition artifacts. Exact formulas are in Table XXII; tracks retained only as reconciled summaries do not support a stronger feature-level claim.

Family	Statistics
Frozen output Z_0	mean entropy, mean confidence, normalized marginal-class entropy
Adapted output Z_a	mean entropy, mean confidence, normalized marginal entropy, frac. confidence > 0.9
Change Z_{change}	entropy drop, marginal-balance drop
Distribution drift	adapted-vs-frozen predicted-class marginal KL
Update behavior	ℓ_2 norm of the adapter parameter update

sits in a known landscape: weighted conformal corrects coverage under known covariate shift [34], adaptive conformal tracks coverage online [36], and beyond exchangeability the coverage gap is bounded by the total-variation drift of the calibration pairs [35], which is conceptually related to but mathematically distinct from our disagreement-region calibration budget β. The certificate's marginal false-adapt control is risk control in the RCPS/Learn-then-Test lineage [38], [39], specialized to the adapt/freeze/abstain loss; PAC prediction sets give set-valued analogues under covariate shift [37].

B. Evidence Map and Benefit Estimator

The evidence map uses only unlabeled batches and model outputs. In the promoted per-condition artifacts retained by this release, it is an 11-dimensional vector computed from the frozen softmax p_0 and the adapted softmax p_a on the adaptation batch: the mean predictive entropy and mean confidence of f_0 and of f_a; the normalized marginal (predicted-class) entropy of each; the entropy and marginal-balance drops between them; the fraction of adapted predictions above confidence 0.9 (a collapse indicator); the KL divergence between the adapted and frozen predicted-class marginals; and the ℓ_2 norm of the adapter's parameter update. An exploratory 16-feature natural-shift panel existed in the development tree, but it is not used to support a promoted result here and is therefore excluded from the method claim. KGA does not explicitly estimate the theoretical margin M; it learns the benefit $\widehat{\Delta} = h_\theta(Z)$ directly from the retained evidence vector Z.

The benefit model predicts $\widehat{\Delta} = h_\theta(Z)$ from labeled development conditions; h_θ is a gradient-boosted regression tree (scikit-learn `GradientBoostingRegressor`, squared error, 250 trees, depth 2, learning rate 0.05, subsample 0.8, seed 0), refit *per dataset and per candidate adapter* rather than shared, so the estimator itself implies no cross-dataset transfer. The complete promoted schema—exact per-feature formulas and families—is given in Appendix B (Table XXII), with per-adapter update hyperparameters in Table XXIII.

Algorithm 1 KGA: Knowability-Guided Adaptation

Require: Frozen f_0; adapter $\mathcal{A}$; batch $X_{1:m}$; $\widehat{\Delta}$, ε, α
Ensure: Decision $g \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$

```
f_a ← A(f_0, X_{1:m})
Z ← φ(X_{1:m}, f_0, f_a)
Δ̂ ← Δ̂(Z);  L ← Δ̂ − ε;  U ← Δ̂ + ε
if L > 0 then
    g ← ADAPT
else if U < 0 then
    g ← FREEZE
else
    g ← ABSTAIN
end if
return g
```

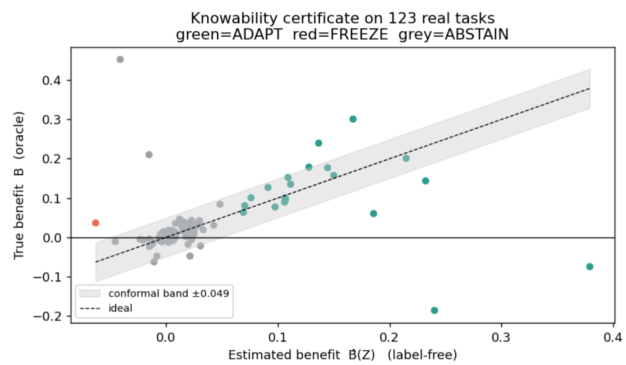


Fig. 3. **The certificate rule in one picture.** KGA estimates the label-free benefit $\widehat{\Delta}(Z)$ and forms an interval $\widehat{\Delta} \pm \varepsilon$: if the interval is entirely positive it adapts, if entirely negative it freezes, and otherwise it abstains. The vertical axis here uses true benefit only for offline audit; deployment decisions use only unlabeled evidence and the pre-calibrated radius.

C. Out-of-Fold Uncertainty Calibration

For calibration condition i, an estimator that did not train on i produces $\widehat{\Delta}_{-i}(Z_i)$ and residual

$$r_i = |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|.$$

The radius is a pre-specified upper residual quantile. On clean development/test splits, ordinary split conformal provides finite-sample marginal coverage under exchangeability. On stress grids, the reported radius uses leave-one-condition-out cross-fitted empirical residual calibration; its observed coverage does not imply the same finite-sample guarantee as split conformal, and jackknife+ is not claimed. We therefore report the calibration design and guarantee level separately for each track.

On the good event $|\widehat{\Delta} - \Delta| \le \varepsilon$, committing adapt only when $L > 0$ implies $\Delta > 0$ (Theorem 3). The implication requires measurable marginal coverage; exchangeability or shift correction is used to justify coverage, while risk alignment is relevant to the population interpretation and estimator transfer. No target labels are used at deployment.

D. Population Frontier versus Empirical Certificate

KGA's commit boundary is not a swept hyperparameter, but the population frontier and the empirical radius must be kept distinct. The *population* frontier of Theorem 2 says a strict action is uniformly sound over $\mathcal{C}_\beta$ iff the population margin $|M|$ exceeds the declared budget β. The realized drift satisfies $|\gamma| \leq \beta$ but is unobserved. KGA instead learns $\widehat{\Delta}$ from Z and uses a finite-sample residual-coverage radius ε, fixed before target-test scoring. The radius neither estimates nor substitutes for β; conditional on valid coverage it controls marginal decision errors without requiring risk alignment. Confidence-thresholding (ATC) and agreement-trend selection (AGL) fit a scalar against a labeled or held-out signal and predict *accuracy*; KGA certifies a strict benefit direction with a calibrated interval and, unlike a committal thresholded estimator, ABSTAINS where its calibrated interval crosses zero—consistent with the structural region studied by Theorem 1, though finite-sample abstentions may also reflect sampling uncertainty, estimator inadequacy, or calibration conservatism. The impossibility construction, not a tuned cutoff, is what separates KGA from "estimate accuracy, then threshold."

E. Selecting and Auditing the Drift Budget

The drift budget β is part of the declared deployment class, not a target-test-tuned hyperparameter. It may be justified from historical domain calibration gaps, a documented domain assumption, or an explicit stress envelope. Sensitivity analysis should report how decisions change over a pre-specified range of budgets. A smaller β asserts stronger calibration transfer and increases commitment; a larger β widens the abstention region. When no credible upper bound is available, the population frontier does not provide a robust sign certificate. Reporting $\beta = 0$ is then only a zero-drift plug-in analysis, not an assumption-free or conservative guarantee.

F. Deployment Semantics and Failure-Safe Behavior

The shadow-state workflow is: (1) checkpoint the active model; (2) propose adaptation on a temporary copy; (3) evaluate frozen and candidate outputs; (4) extract Z; (5) compute $\widehat{\Delta} \pm \varepsilon$; (6) commit or roll back; and (7) log the configuration, evidence, interval, and action. Abstention applies to the adaptation update, not to prediction. ADAPT commits the candidate; FREEZE rolls it back; and ABSTAIN retains the frozen state while recording that evidence was insufficient rather than certifiably harmful. The implementation should automatically freeze or abstain when evidence is missing, the batch is too small, a checkpoint or adapter configuration differs from calibration, the evidence lies outside calibration support, numerical values are invalid, or no valid radius is available. In the reference implementation (`kbound_pkg`), the gate is applied per batch (`KGA.decide_from_batch`); on ABSTAIN or FREEZE the adaptation optimizer scales the candidate update by `abstain_scale`=0, i.e. the update is skipped and the frozen state f_0 is served—this is the rollback mechanism. The failure-safe defaults instantiate the diagnostics of Table XX: abstain when the batch is smaller than the calibration batch size, when Z lies outside calibration support, when a checkpoint or adapter-configuration hash differs from the calibrated one, or when the radius or evidence is missing or numerically invalid. In continual operation the last certified state replaces f_0 as the fallback.

TABLE VI
KGA CONTROLLER *ADDED* COST (RESNET-18/CIFAR, APPLE SILICON, PYTHON 3.12.13; MEASURED VIA `SCRIPTS/RECOMPUTE/COST_PROFILE.PY`). TIMINGS ARE ENVIRONMENT-SPECIFIC AND EXCLUDE ADAPTATION/INFERENCE.

Component	Cost
Evidence extraction (per batch)	0.098 ms
Benefit-model evaluation (per decision)	0.245 ms
Controller added latency	**0.343 ms**
Rollback checkpoint (on-disk size)	44.77 MB
Serialized benefit model	194.9 KB

G. Computational Cost

KGA cannot avoid the cost of proposing an update because candidate adaptation includes forward and backward computation before commitment. A complete profile must separately report frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit-estimator evaluation, model-copy memory, rollback, and end-to-end window latency. The controller's *added* cost is measured in Table VI (`scripts/recompute/cost_profile.py`; artifact `experiments/kbound/results/controller_cost_v1/cost_profile.json`): evidence extraction 0.098 ms and benefit-model evaluation 0.245 ms per decision (0.343 ms total) on the reported Apple-silicon environment. The referenced ResNet-18/CIFAR checkpoint is 44.77 MB on disk and the serialized benefit model is 194.9 KB. Candidate adaptation and inference are unchanged by KGA—f_a is evaluated regardless—and dominate end-to-end latency.

VI. EXPERIMENTAL SETUP

A. Research Questions

The experiments address seven questions: **RQ1** whether the measured commit transition follows the theoretical frontier; **RQ2** whether the interval controls marginal false adaptation; **RQ3** whether KGA beats both fixed policies in mixed detectable regimes; **RQ4** whether it prevents damage on held-out natural shifts; **RQ5** whether it improves over simple and neighboring label-free gates; **RQ6** which evidence and calibration choices matter; and **RQ7** what computational overhead the controller adds.

B. Datasets and Shift Taxonomy

The evaluation separates controlled corruption/composition stress, large-scale corruption, natural institutional or geographic shifts, domain generalization, and low-margin diagnostic transfers. This taxonomy matters because a router can beat both fixed policies only when the optimal action varies across conditions and that variation is detectable from label-free evidence. The nine tracks draw on CIFAR-10-C [21], ImageNet [20], the WILDS Camelyon17, iWildCam, and RxRx1 shifts [22], [23], [24], Office-Home [25], PACS via DomainBed [26], and CIFAR-10.1 [28]; models build on RobustBench checkpoints [27] with PyTorch [29] and scikit-learn [30].

C. Models, Adapters, and Leakage Control

For every track, the source checkpoint, candidate adapter, update mode, learning rate, number of steps, batch construction, adapted parameter subset, and random seeds must be declared before held-out scoring. Adapter selection, evidence selection, benefit-model fitting, and residual calibration use development data only. All choices are frozen before the target test cells are scored once; target labels are opened afterward solely for offline audit. The calibration and scoring configuration is uniform across tracks except where noted in Table VIII.

Exact per-adapter update settings (optimizer, steps, learning rate, and the mild/aggressive operating points) are tabulated per track in Appendix B, Table XXIII.

D. Baselines

We distinguish fixed policies (always-freeze and always-adapt), an offline per-condition oracle, simple label-free gates (confidence, entropy, drift/KL, and ATC-style), KGA without its radius, and neighboring methods or ports inspired by POEM and AETTA. Each comparison must state whether the baseline is an official implementation, a faithful reimplementation, or a protocol-matched port.

E. Metrics and Statistical Testing

The primary decision metrics are regret-to-oracle, marginal false-adapt $\mathrm{FA_u} = \mathbb{P}(\textsc{adapt}, \Delta \leq 0)$, conditional false-adapt $\mathrm{FA_c} = \mathbb{P}(\Delta \leq 0 \mid \textsc{adapt})$, adapt rate, and decision coverage $\mathbb{P}(\textsc{adapt or freeze})$. Accuracy or AUROC is reported as the task metric. The theorem controls $\mathrm{FA_u}$ under its assumptions; $\mathrm{FA_c}$ is descriptive. A beats-both verdict requires lower regret than both fixed policies under the declared statistical criterion and $\mathrm{FA_u} \leq \alpha$. Paired bootstrap confidence intervals, their resampling unit, seed aggregation, and Holm correction are declared per comparison family.

F. Evidence-Status Policy

Locked results trace to raw cells/seeds and a frozen aggregate; **reconciled** results correct a prior interpretation using raw records; **provisional** summaries require canonical replay; and **diagnostic** runs are complete but deliberately not promoted. Only locked and reconciled rows support the empirical headline.

G. Hardware and Software

Runs use Python 3.12, PyTorch 2.5.1, and scikit-learn for the benefit estimator. Accelerator and checkpoint details are recorded per run manifest rather than pooled into one hardware claim: CIFAR experiments used the saved ResNet-18 setup; the authoritative ImageNet-C result used ResNet-50 on seed 0. Hardware affects runtime but not the decision definitions or replayed metrics.

VII. Results

The results are organized by regime rather than by dataset prestige. Controlled mixed regimes test policy improvement, one-sided natural shifts test no-harm behavior, and low-margin transfers test whether the method declines unsupported claims.

A. Synthetic validation of the frontier

Before the real-data tracks we validate the theory directly on synthetic ground truth, where the observable margin M, the unobserved drift $\gamma \sim \mathrm{U}(-\beta, \beta)$, and hence the true benefit sign $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ are all controlled. In the real-data tracks that follow, β is *not* numerically supplied to KGA: the empirical decision uses $\widehat{\Delta} \pm \varepsilon$, and the β-frontier is evaluated directly only in this synthetic study. Evidence is fed to the reported cross-fitted rule (leave-one-out gradient-boosted $\widehat{\Delta}$ with an empirical residual radius, §V); nothing is hand-tuned. Three predictions hold (Fig. 5; α=0.10). **(i) Recovery:** $\widehat{\Delta}$ tracks Δ at 90.0% empirical coverage, matching the nominal target in this simulation. In this construction unresolved drift dominates the benefit-estimation residual, causing the calibrated radius to approximately track β. **(ii) Frontier** (Thm 2): the commit rate jumps from 6.5% for $|M| < \beta$ to 96.7% for $|M| > \beta$—a transition *at* the predicted boundary—and the committed sign is correct 99.3% of the time. **(iii) Certificate** (Thm 3): the unconditional false-adapt rate stays under budget for every $\beta \in \{0.05, \ldots, 0.20\}$ ($\mathrm{FA_u} \leq 0.014 \ll \alpha$). The frontier is not imposed; it appears because the synthetic residual scale is controlled by the deliberately hidden drift.

B. CIFAR-10-C stress grid (5 seeds)

The per-corruption suite is helpful-dominated; the *stress* grid crosses severity, batch size, composition, and update aggressiveness (432 conditions/method, ResNet-18; pre-registered as Protocol A; seeds 0–4). Harmful base rates are 34% (Tent) and 27% (EATA). KGA **beats both** trivial policies for Tent and EATA at 0% false-adapt (Table IX). Among harmful cells KGA adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains. For both Tent and EATA, the regret gaps to *both* fixed policies are positive with 95% bootstrap CIs excluding zero, Holm-corrected over the comparison family,

TABLE VII
DATASET ROLES IN THE EVALUATION. THE EXPECTED REGIME IS A HYPOTHESIS FIXED BY THE PROTOCOL, NOT A CONCLUSION INFERRED FROM THE FINAL SCORE.

Track	Shift type	Candidate role	Evaluation purpose
CIFAR-10-C stress	corruption/composition/update stress	Tent/EATA/SAR	mixed detectable decision benchmark
ImageNet-C noise	large-scale corruption	Tent/EATA/SAR	scale and candidate dependence
Camelyon17	hospital/institution shift	EATA	helpful-dominated natural safety
iWildCam	geographic camera shift	Tent	harmful-dominated natural safety
Office-Home	visual domain/style shift	SAR	held-out domain safety
RxRx1	experimental/laboratory shift	online adapter	harmful-dominated natural safety
PACS	domain generalization	fixed candidate	breadth and null behavior
ImageNet-R	rendition shift	candidate panel	weak-evidence diagnostic
CIFAR-10.1	natural resampling	TTA candidates	low-margin transfer diagnostic

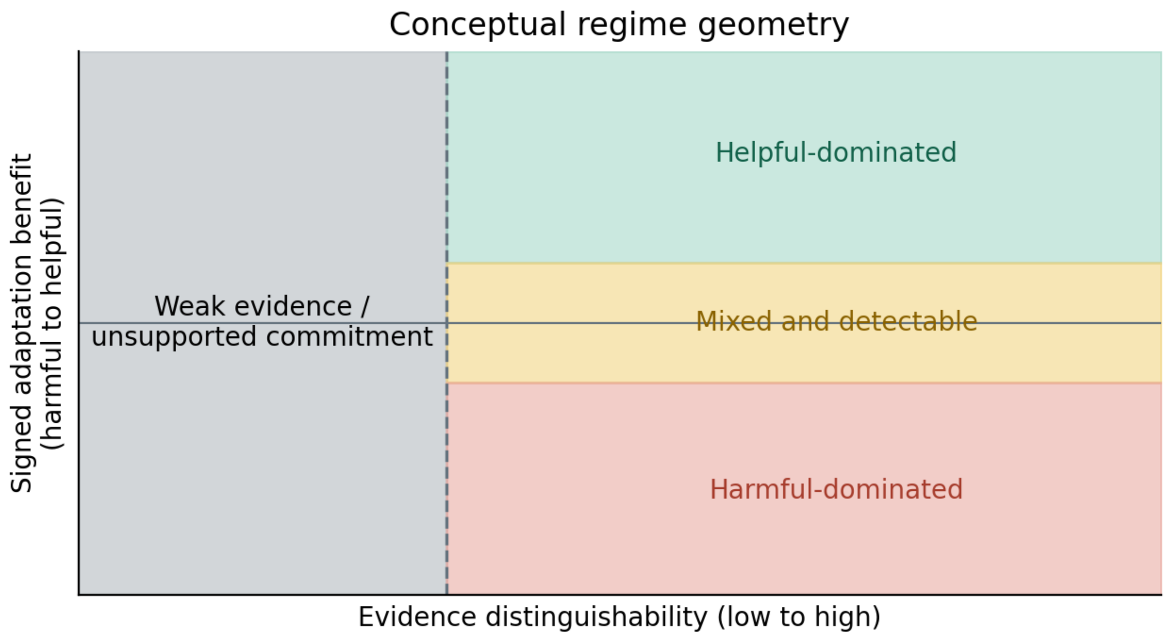


Fig. 4. **Regime geometry for K-Bound.** The important axis is not dataset name but where the deployment sits relative to the frontier. Helpful- or harmful-dominated regimes usually produce no-harm ties with the better fixed policy; low-margin regimes force abstention; mixed and detectable regimes are where a valid adapt/freeze/abstain router can beat both fixed policies. This is a *conceptual schematic*: axis positions are illustrative regime categories, not measured coordinates (the actual per-track verdicts are in Table I).

at false-adapt 0/2160.[1] The saved SAR aggregate is not promoted: a later seed-0 replacement causes the current raw tree to replay different SAR fixed-policy regrets, so SAR coverage and verdict require a clean five-seed rebuild. The Pareto panel in Fig. 6 is retained as archived decision geometry; the promoted result is limited to Tent and EATA.

[1]Design-based intervals, resampling the pre-registered mixing-ratio distribution of the stress-grid design (Protocol A; the 11-point Pareto over harmful fraction, N_{boot}=5000): Tent +0.019 [0.011, 0.026] vs. always-adapt, +0.080 [0.052, 0.107] vs. always-freeze; EATA +0.015 [0.009, 0.021] and +0.075 [0.049, 0.100]. A conventional paired per-condition bootstrap over the 432 realized conditions (`scripts/recompute/percondition_bootstrap.py`) returns the same Tent/EATA verdict.

C. Decision-baseline comparison: empirical safety of the calibrated certificate

KGA's claim goes beyond "better than always-adapt/always-freeze": it is better than *simple label-free decision rules* under the same locked protocol. On the CIFAR-10-C stress grid (Tent candidate, identical cells), we score six rules from the same evidence (Table X): a confidence gate (adapt iff adaptation raised mean confidence), an entropy gate (adapt iff it lowered entropy—Tent's own objective), a drift/KL threshold, an ATC-style accuracy-prediction gate, KGA *without* its conformal radius (adapt iff $\widehat{\Delta} > 0$), and the full certificate. Only KGA is derived from a coverage-based false-adapt certificate; on this stress grid the leave-one-cell-out radius provides strong *empirical* coverage, not the exact distribution-free validity of clean split conformal or jackknife+. It attains zero observed

TABLE VIII

Per-track calibration and scoring configuration (α=0.10 throughout; benefit estimator family and calibration recipe are used across tracks, with parameters refit per dataset and candidate adapter, §V). Backbones: ResNet-18 (CIFAR), ResNet-50 (ImageNet-C); the natural-shift backbones follow the WILDS/DomainBed defaults per track.

Track	Calibration unit	Radius rule	Test unit
CIFAR-10-C stress	grid cell (leave-one-cell-out)	LOO jackknife $q_{0.9}$	5×432 cells
ImageNet-C	grid cell (leave-one-cell-out)	LOO jackknife $q_{0.9}$	27 cells
PACS (LODO)	source-domain cells	LOO jackknife $q_{0.9}$	108 cells/target
Camelyon17	dev $\{0,1\}$	archived OOF empirical radius	test $\{2,3,4\}$
iWildCam H v2	calibration $\{0\}$	archived OOF empirical radius	held-out $\{1\}$
Office-Home M v2	calibration $\{0,1\}$	archived OOF empirical radius	target test $\{0,1\}$
RxRx1 J	development $\{0,\dots,4\}$	archived OOF empirical radius	test $\{5,\dots,9\}$

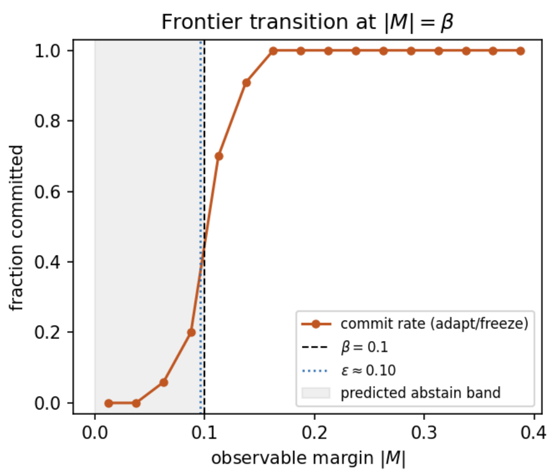


Fig. 5. **Measured benefit-sign frontier** (synthetic ground truth, archived cross-fitted rule). The commit rate transitions sharply from abstention to adapt/freeze near $|M|=\beta$; the calibrated radius ε (dotted) approximately tracks β in this construction. Companion panels—estimator recovery and $\mathrm{FA_u} \le \alpha$ across β—in `fig_frontier_recovery.png`, `fig_frontier_fa_coverage.png`.

$\mathrm{FA_u}$ on this grid while staying near-oracle on regret (the radius-free variant keeps $\mathrm{FA_u}$=0.049<α empirically, but with no such guarantee). The confidence/entropy gates false-adapt on ∼74% of harmful cells; the drift threshold, tuned to the budget, degenerates to always-freeze. The KGA-without-radius row isolates the radius: removing it lowers regret slightly (0.0017 → 0.0004) but raises $\mathrm{FA_u}$ from 0 to 0.049 (to 0.141 on harmful cells): the radius is what turns a good benefit *estimate* into a false-adapt *certificate*. The archived values and a clean exact-rank recomputation are stored under `experiments/kbound/results/gate_baselines_v1/`; the recomputation script

candidate	policy	mean acc	regret↓	$\mathrm{FA_u}$	coverage	beats both?
Tent	always-adapt	0.786	0.0079	—	1.00	
	always-freeze	0.671	0.1241	—	1.00	
	K-Bound	**0.793**	**0.0016**	**0**	0.67	**yes**
EATA	always-adapt	0.798	0.0033	—	1.00	
	always-freeze	0.671	0.1314	—	1.00	
	K-Bound	**0.801**	**0.0013**	**0**	0.63	**yes**

TABLE IX

CIFAR-10-C stress grid (432 conditions/method, five seeds, α=0.1, Protocol A). "coverage" is decision coverage $\mathbb{P}$[adapt ∨ freeze]; both fixed policies commit on every cell. SAR is excluded because the current raw tree no longer replays its archived aggregate after a seed-0 replacement.

TABLE X

Decision-baseline comparison, CIFAR-10-C stress grid

(Tent candidate, n=432 cells, 149 harmful, α=0.1). Lower regret and lower $\mathrm{FA_u}$ are better. Only the conformal certificate keeps $\mathrm{FA_u}$=0 on the full grid and on harmful cells while staying near-oracle.

Decision rule	regret↓	$\mathrm{FA_u}$	$\mathrm{FA_c}$	adapt	cov.	$\mathrm{FA_u}$(harm)
confidence gate	0.0084	0.257	0.301	0.85	1.00	0.745
entropy gate	0.0086	0.255	0.304	0.84	1.00	0.738
drift/KL gate	0.1232	0.000	0.000	0.00	1.00	0.000†
ATC-style gate	0.0045	0.116	0.172	0.67	1.00	0.336
KGA (no radius)	0.0004	0.049	0.071	0.68	1.00	0.141
KGA (certificate)	0.0017	**0.000**	**0.000**	0.51	0.68	**0.000**

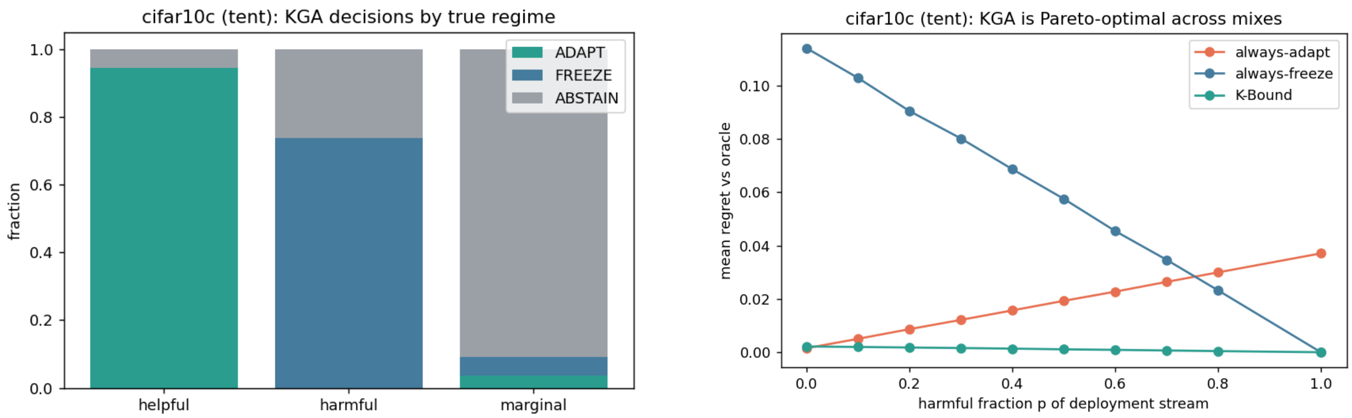
†Degenerate: the budget-tuned drift gate never adapts (adapt-rate 0), so its "0 false-adapt" is just always-freeze (regret 0.123).

is `scripts/recompute/gate_baseline_comparison.py`.

D. Mixed head-to-head vs. POEM-/AETTA-style baselines (pre-registered)

The stress-grid wins above beat the *trivial* policies always-adapt and always-freeze. A harder question is whether KGA improves over strong protocol-matched label-free decision baselines inspired by POEM [10] and AETTA [15], on a *mixed* harmful+helpful stream. We pre-registered the benchmark, metrics, and WIN/TIE/LOSE criterion *before* computing any number (`MIXED_BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals Z and benefit B from the locked CIFAR-10-C stress grid (Tent, 432 conditions/seed, five seeds; harmful fraction ≈33%, i.e. the always-adapt $\mathrm{FA_u}$ of Table XI); only the decision rule differs. The competitors are *protocol-matched POEM-style and AETTA-style baselines*: ports of the published protectors onto this protocol with documented simplifications (batch-summary entropy rather than per-sample streams). Official per-sample POEM and dropout-based AETTA reproduction remains a camera-ready faithfulness check.

Verdict: WIN. KGA lowers mean regret-to-oracle vs. both ported baselines with paired-bootstrap 95% CIs entirely below zero and Holm correction (α=0.05), at $\mathrm{FA_u}$=0 ≤ α (Table XI). The ported POEM-/AETTA-style baselines false-adapt at ∼24–25% on the same stream, consistent with the absence of an explicit marginal false-adapt certificate under this ported protocol. Secondary pooled sets (Tent+EATA, EATA



(a) decisions by true regime

(b) regret as harmful fraction changes

Fig. 6. **CIFAR-10-C stress grid is a decision result, not only an accuracy result.** KGA adapts on helpful cells, freezes harmful cells, abstains on marginal cells, and therefore lies on the regret front: it ties always-adapt at p=0 and beats both fixed policies once harmful mass is nontrivial.

TABLE XI

PRE-REGISTERED MIXED HEAD-TO-HEAD ON THE LOCKED CIFAR-10-C STRESS STREAM. POEM/AETTA ROWS ARE PROTOCOL-MATCHED STYLE PORTS, NOT OFFICIAL REPRODUCTIONS. ALL POLICIES USE THE SAME HELD-OUT CELLS, CANDIDATE ADAPTER, LOGGED LABEL-FREE EVIDENCE, REGRET-TO-ORACLE METRIC, AND UNCONDITIONAL FALSE-ADAPT METRIC. THE COMPARISON TESTS KGA AGAINST STRONG LABEL-FREE DECISION GATES UNDER THIS LOCKED PROTOCOL, NOT AGAINST FULL OFFICIAL POEM/AETTA IMPLEMENTATIONS.

policy	mean regret↓	FA_u	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	−0.0064	yes
always-freeze	0.1241	0.000	1.00	−0.1225	yes
POEM-style	0.0088	0.245	1.00	−0.0072 [−0.0088, −0.0057]	yes
AETTA-style	0.0073	0.240	1.00	−0.0058 [−0.0071, −0.0044]	yes
KGA	**0.0016**	**0.0000**	0.6847	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

alone) yield the same WIN verdict (`experiments/kbound/results/mixed_headtohead_v1/`). A separate recomputation pipeline (paired bootstrap 10^4, pre-registered) replicates all three configurations, with both regret-gap CIs below zero in each and FA_{KGA}=0 versus 11–33% for the competitors (`research_lock/WIN_HUNT_v3_ARM_F_result.json`).

a) Baseline faithfulness.: The POEM and AETTA rows in Table XI are protocol-matched ports, not official reproductions. They are included to test whether KGA's conformal benefit-sign certificate improves over strong label-free committal gates under the same logged evidence, candidate adapter, held-out cells, regret metric, and false-adapt metric (Table XII). They do not claim to reproduce POEM's full per-sample betting process or AETTA's full dropout-based estimator. Official-code reproduction is left as an additional faithfulness check; the present claim is therefore limited to the locked protocol-matched comparison.

E. ImageNet-C SAR (mechanism-faithful)

On ImageNet-C noise (ResNet-50, 27 cells: three noise corruptions × severities $\{1, 3, 5\}$ × three batch-composition regimes (iid, imbalanced, single-class), seed 0, full-scale), SAR is harmful on 14.8% of cells under a mechanism-faithful re-implementation (SAM, recovery reset, EMA) at the learning rate shared across candidates. Always-adapt drops to 0.379 accuracy while KGA reaches 0.431, **beating both** trivial policies (regret 0.0108 vs 0.0625/0.0319). The gap to the better fixed policy (always-freeze) has a 95% paired-bootstrap CI of $[-0.032, -0.011]$; the gap to always-adapt is $[-0.115, -0.001]$. Both intervals are regenerated from the saved 27 cells, so ImageNet-C SAR is a paired-bootstrap beats-both rather than a point estimate. No multiple-comparison correction is claimed for this isolated table. EATA stays helpful-dominated (KGA ties always-adapt); on Tent the harmful regime now dominates (63% of cells) and KGA abstains to always-freeze (Table XIII). Under the same logged conditions, an ATC-style confidence gate false-adapts on the harmful SAR cells—it adapts wherever confidence is high, which is exactly where SAR collapses—while KGA abstains on them, so the certificate beats not only the two fixed policies but a simple label-free decision gate under the same protocol. The result uses our mechanism-faithful SAR reimplementation at a shared learning rate; SAR's official gentler schedule is left as a robustness check, so the claim is about this harmful-SAR operating point rather than all SAR configurations.

F. Natural Shifts and Consolidated Summary

All benchmark tracks use the same decision target: the adaptation benefit $\Delta = R(f_0) - R(f_a)$, evaluated against the same per-condition oracle, always-adapt, and always-freeze policies. The candidate adapter is locked using development data. Evidence Z, the benefit estimator $\widehat{\Delta}$, and radius ε are then frozen before scoring the declared test cells; test labels are used only after decisions for offline regret and false-adapt audit. Stress grids use leave-one-cell-out residuals; semantic domain-shift tracks use a dev/test

TABLE XII
Baseline faithfulness audit for POEM/AETTA comparisons.

Baseline	Original method needs	Our locked protocol uses	What is matched	What is not claimed
POEM-style	online entropy/betting monitor over adaptation stream	batch-summary entropy and logged per-condition evidence	same held-out cells, same candidate, same false-adapt/regret metrics	not official per-sample POEM reproduction
AETTA-style	dropout/disagreement-based label-free accuracy estimation	protocol-matched accuracy/benefit gate from logged evidence	same evidence budget, same decision task, same held-out stream	not official dropout-AETTA reproduction
KGA	benefit estimator + conformal radius	out-of-fold $\widehat{\Delta}$ and ε	same stream, same oracle/regret/$\mathrm{FA_u}$ metric	official method in this paper

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.399	0.0216	63%	
	always-freeze	0.410	0.0114		
	K-Bound	**0.410**	**0.0114**		no (ties freeze)
EATA	always-adapt	**0.443**	0.0000	0%	no (ties adapt)
	always-freeze	0.410	0.0336		
	K-Bound	**0.443**	0.0000		
SAR	always-adapt	0.379	0.0625	15%	
	always-freeze	0.410	0.0319		
	K-Bound	**0.431**	**0.0108**		**yes**

TABLE XIII
ImageNet-C, mechanism-faithful SAR (27 cells: 3 noise corruptions, ResNet-50, α=0.1); full-scale run (2026-07-09) superseding the earlier 36-cell provisional configuration.

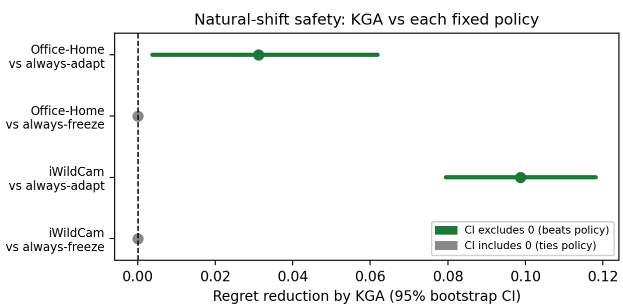


Fig. 7. **Natural-shift no-harm, not beats-both.** Positive values favor KGA. Office-Home and iWildCam beat always-adapt but tie always-freeze under the corrected out-of-fold replay; the intervals are paired held-out-condition bootstrap intervals from `research_lock/KBOUND_WIN_BOOTSTRAP_CIS_oof.json`.

split. We report $\mathrm{FA_u} = \mathbb{P}[\textsc{adapt}, \Delta \leq 0]$ and do not interpret $\mathrm{FA_c}$ as a certificate.

a) Evidence tiers.: **Locked** means the table row traces to raw cells/seeds and a locked aggregate; **reconciled** means a prior pooled interpretation was corrected from its raw records; **provisional** means a derived out-of-fold summary exists but does not reproduce the nearby canonical scorer; **diagnostic** means a completed run that is deliberately not promoted. Only locked and reconciled rows support the empirical headline.

b) Multi-seed stability (Camelyon17).: The one natural-shift track with per-condition logs at multiple seeds is Camelyon17 (4 seeds, 9 composition-stress conditions/seed). Aggregating the deployed per-seed decisions (`scripts/recompute/multiseed_natural.py`; locked artifacts under `experiments/kbound/results/camelyon17_multiseed_v1/`) shows the no-harm outcome is *candidate-dependent* and stable only for the milder adapter (Table XIV): with Tent, KGA is **stable no-harm** across all four seeds—it ties always-freeze, beats always-adapt, at $\mathrm{FA_u}$=0 every seed; with EATA the margin is inconclusive; and with the aggressive helpful-dominated SAR arm KGA over-freezes (worse than always-adapt, $\mathrm{FA_u}$ up to 0.11), consistent with the candidate dependence seen elsewhere. Per-seed multi-seed replays for iWildCam, Office-Home, and RxRx1 are single-run in the current artifacts and remain future work.

TABLE XIV
Multi-seed no-harm (Camelyon17, seeds 0–3). Regret mean±std across seeds; $\mathrm{FA_u}$ is the per-seed maximum. Reproduced by `scripts/recompute/multiseed_natural.py` from the committed per-seed logs.

candidate	KGA regret	adapt	freeze	$\mathrm{FA_u}$	verdict
Tent	0.020±0.023	0.138	0.020	0.00	stable no-harm
EATA	0.039±0.025	0.042	0.042	0.00	inconclusive
SAR	0.041±0.017	0.000	0.065	0.11	over-freezes (helpful-dom.)

G. Weak-Evidence and Negative Regimes

ImageNet-R and CIFAR-10.1 remain diagnostics. Their outcomes are consistent with weak evidence, low margin, estimator inadequacy, or calibration-transfer failure; they do not by themselves prove structural non-identifiability. PACS is also incomplete relative to its three-seed registration.

H. Physical-Study Pre-registration

We preregister a two-phone physical package-inspection study with distinct source, calibration, held-out, and replication sessions. The protocol, leakage checks, and outcome-free result templates are provided in Appendix F. No physical result is claimed until fresh sessions pass the repository's fail-closed publication gate; mock or reconstructed clips are never admissible evidence.

I. Constructed Heterogeneous Routing

The three-source OOF mixture combines separately calibrated Office-Home, iWildCam, and Camelyon17 conditions. It demonstrates routing value under a researcher-

TABLE XV
Uniform nine-track empirical panel. Regrets are ordered as KGA / always-adapt / always-freeze; lower is better. BB = beats both fixed policies. The final column is part of the result: it prevents a point estimate, a reconstructed summary, and a locked multi-seed result from being presented as equivalent evidence.

Track	Evaluation unit	Result	Decision interpretation	Evidence tier
CIFAR-10-C stress	archived 5×432 aggregate	Tent: 0.0016/0.0079/0.1241; EATA: 0.0013/0.0033/0.1314; $\mathrm{FA_u} = 0$	Tent/EATA CI BB; SAR withheld	archived aggregate; raw replay mismatch
ImageNet-C SAR	27 cells (3 noise corr.)	0.0108/0.0625/0.0319; $\mathrm{FA_u} = 0$	**paired-bootstrap BB**; 12/27 abstain	source-backed single seed
Camelyon17 OOD	genuine OOD test $n = 18$ (support $n = 36$)	0.0000/0.0000/0.1381; $\mathrm{FA_u} = 0$	ties adapt; no BB	reconciled raw records
iWildCam H v2	held-out test $n = 72$	0.0041/0.1028/0.0041; $\mathrm{FA_u} = 0$	no-harm, ties freeze	reconciled (OOF lock)
Office-Home M v2	target test $n = 35$	0.0157/0.0468/0.0158; $\mathrm{FA_u} = 0$	no-harm, tiny point edge	reconciled (OOF lock)
RxRx1 J	OOD test $n = 60$	0.0000/0.2531/0.0000; $\mathrm{FA_u} = 0$	ties freeze; no BB	locked raw run
PACS	four LODO targets, 108 cells each	three safety targets ($\mathrm{FA_u}$=0); one null (photo, $\mathrm{FA_u}$=0.056)	no promoted BB	diagnostic; 1 of 3 planned seeds
ImageNet-R D	3 seeds ×10 backbones	no CI-robust BB; one point-only arm fails the second CI	abstention / fixed-policy ties	diagnostic; 3 of 4 planned seeds
CIFAR-10.1 K	cross-seed test $n = 48$	0.0021/0.0190/0.0017; $\mathrm{FA_u} = 0.167$, $\mathrm{FA_c} = 0.444$	fails transfer bar; no claim	diagnostic; corrected metric label

TABLE XVI
Primary numeric evidence used for empirical claims. The three-source row is a constructed routing mixture, not a natural-transfer result.

Track	Unit	KGA	Adapt	Freeze	Status
CIFAR-10-C Tent	5×432	.0016	.0079	.1241	CI BB, $\mathrm{FA_u} = 0$
CIFAR-10-C EATA	5×432	.0013	.0033	.1314	CI BB, $\mathrm{FA_u} = 0$
ImageNet-C SAR	27 cells	.0108	.0625	.0319	paired-bootstrap BB, $\mathrm{FA_u} = 0$
Camelyon17 OOD	$n = 18$	.0000	.0000	.1381	reconciled no-harm
RxRx1 J	$n = 60$	.0000	.2531	.0000	locked no-harm
Three-source OOF mixture	$n = 143$	.0059	.0632	.0342	CI BB (constructed); locked

constructed heterogeneous stream, not single-dataset natural-shift dominance or transfer to unseen shift types.

J. Consolidated Claim and Guarantee Accounting

Statement	Status	Requirement
Abstention needed in matched-evidence opposite-benefit worlds	theorem	stated model class
$\mathrm{FA}_u \le \alpha$ (unconditional false-adapt)	theorem	measurable marginal coverage; exchangeability/shift correction only justifies coverage
$\mathrm{FA}_c \le \alpha$ (conditional false-adapt)	**not claimed**	report FA_c descriptively only
Nine-track panel	**audit**	locked, reconciled, or provisional tiers shown explicitly
Stress-grid beats-both (Tent/EATA)	empirical	locked Protocol A v1 stress grid
ImageNet-C SAR paired-bootstrap beats-both	empirical	source-backed 27-cell noise grid; both paired CIs exclude 0
Reconciled natural safety (Camelyon17, RxRx1)	empirical	raw OOD/test artifacts
Office-Home and iWildCam OOF summaries	reconciled	OOF bootstrap lock
Three-source mixed routing aggregate	empirical	constructed OOF mixture, not transfer
Universal improvement	**not claimed**	impossible in general
ε estimates drift budget β	**false**	β is declared; ε is empirical

VIII. Ablations and Sensitivity

A. Value of the Uncertainty Radius

Table X isolates the radius: KGA without the radius has lower point regret but a higher empirical false-adapt rate, whereas the full certificate sacrifices coverage to eliminate observed harmful commitments on the locked grid. This comparison separates benefit-prediction quality from certification.

B. Alpha, Evidence, Estimator, and Transfer Sensitivity

All ablations reuse the logged CIFAR-10-C stress evidence (per-condition Z and true benefit B; 432 conditions/candidate, seed 0) and an *exact-rank* empirical residual radius $\varepsilon = r_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$ (`scripts/ablation_exactrank.py`; recomputed artifact

TABLE XVII

ABLATION (I): SAFETY-LEVEL α SWEEP (EXACT-RANK EMPIRICAL RADIUS), CIFAR-10-C TENT (n=432). FA_u=0 AT EVERY TESTED α; THE RADIUS-FREE ROW INCREASES IT.

α	regret↓	FA_u	adapt	coverage
0.01	0.0036	0.000	0.45	0.48
0.05	0.0021	0.000	0.49	0.63
0.10	0.0017	0.000	0.51	0.68
0.20	0.0011	0.000	0.53	0.74
no radius	0.0004	0.051	0.68	1.00

TABLE XVIII

ABLATION (II): BENEFIT ESTIMATOR (EXACT-RANK EMPIRICAL RADIUS), CIFAR-10-C TENT (α=0.10). OBSERVED FA_u REMAINS NEAR ZERO ACROSS ESTIMATORS.

estimator	regret↓	FA_u	adapt	coverage
gradient boosting	0.0017	0.000	0.51	0.68
random forest	0.0017	0.002	0.51	0.65
ridge (linear)	0.0043	0.000	0.45	0.47
small MLP	0.0143	0.000	0.36	0.36

`ablation_exactrank.json`, with full input hashes and software versions). The eight-fold out-of-fold residual procedure is cross-fitted empirical calibration, not exact split-conformal coverage. At Tent α=0.10, the rerun gives regret 0.0017, FA_u=0, adapt rate 0.51, and decision coverage 0.68.

(i) Safety level α. Sweeping α (Table XVII) trades decision coverage for regret while holding observed FA_u=0 at every tested level; the radius-free variant increases observed FA_u to 0.051. **(ii) Estimator.** Swapping the benefit regressor (Table XVIII) leaves the empirical safety pattern nearly intact: gradient boosting, random forest, and ridge all keep $FA_u \leq 0.002$; a small MLP over-abstains on this small calibration set. **(iii) Evidence family.** Dropping any single evidence family (frozen, adapted, change, drift, or update) keeps regret in $[0.0013, 0.0019]$ at FA_u=0: no single family is load-bearing. **(iv) Cross-adapter transfer.** Fitting the benefit model on one adapter and applying it to another (Table XIX) exceeds the empirical α target in most directions (SAR→Tent $FA_u = 0.25$), which is exactly why the deployed method refits per adapter. **(v) Drift budget β.** The population-frontier β sweep is the synthetic ground-truth validation of §VII-A, kept distinct from the empirical radius ε.

IX. DISCUSSION AND FAILURE MODES

A. Why Natural Shifts Mainly Produce No-Harm

A single natural benchmark is often one-sided: adaptation helps almost everywhere or harms almost everywhere. In the first case, always-adapt is already near-oracle; in the second, always-freeze is near-oracle. A three-way router gains a strict margin over both fixed policies only when helpful and harmful conditions coexist and the evidence distinguishes them. Natural no-harm results and mixed-regime wins are therefore different operating points of the same decision geometry.

TABLE XIX

ABLATION (IV): CROSS-ADAPTER TRANSFER (EXACT-RANK EMPIRICAL RADIUS, α=0.10). TRANSFERRING ONE BENEFIT MODEL ACROSS ADAPTERS EXCEEDS THE EMPIRICAL FA_u TARGET IN MOST DIRECTIONS; PER-ADAPTER CALIBRATION IS REQUIRED.

fit → apply	regret↓	FA_u	adapt	coverage
Tent → EATA	0.0021	0.000	0.54	0.63
EATA → Tent	0.0010	0.007	0.55	0.69
Tent → SAR	0.0173	0.125	0.50	0.72
EATA → SAR	0.0236	0.192	0.57	0.62
SAR → Tent	0.0085	0.255	0.84	0.84
SAR → EATA	0.0027	0.116	0.75	0.75

TABLE XX

DEPLOYMENT FAILURES, DIAGNOSTICS, AND SAFE RESPONSES.

Failure	Diagnostic	Safe response
Calibration transfer fails	support/residual audit	recalibrate, widen radius, or abstain
True drift exceeds β	external domain monitoring	revise class/budget; freeze meanwhile
Evidence out of support	distance or density check in Z-space	abstain
Adapter/checkpoint changed	configuration hash mismatch	require recalibration
Batch too small	minimum-size rule	wait for more data or freeze
Continual state accumulates error	checkpoint/stream monitor	rollback to last certified state
Missing/invalid evidence	schema and numerical validation	freeze

B. Four Sources of Abstention

Not every abstention has the same interpretation. *Structural non-identifiability* means opposite-benefit worlds share the same admissible evidence law. *Finite-sample uncertainty* means the sign is identifiable in principle but the interval still crosses zero. *Estimator inadequacy* means the chosen model fails to extract an available signal. *Calibration failure* means a development relationship does not transfer to deployment. We distinguish these causes throughout the evaluation rather than treating every null result as evidence of structural unknowability.

C. Operational Failure Modes

Table XX maps each deployment failure to a diagnostic and a safe response.

D. When to Deploy K-Bound

K-Bound is most valuable when harmful updates are plausible, the cost of a bad commit is high, a frozen fallback is available, and historical or structural information supports calibration. It is less useful when adaptation is uniformly benign, when no credible drift class or calibration set exists, or when the candidate update cannot be evaluated and rolled back safely.

X. LIMITATIONS

Scope of empirical wins. On the stress grid KGA beats both for Tent/EATA while SAR is withheld pending a clean replay (Table IX); on ImageNet-C the pattern flips by candidate—Tent/EATA are robust while SAR collapses and KGA beats both under a mechanism-faithful re-run (Table XIII). Per-corruption CIFAR-10-C and helpful natural shifts are insurance regimes, not policy wins.

Real-shift scope. The uniform panel deliberately distinguishes three outcomes. Camelyon17 genuine OOD and RxRx1 are reconciled/locked no-harm results: KGA ties the better fixed policy at zero observed false-adapt. Office-Home and iWildCam are reconciled from their out-of-fold bootstrap lock (`research_lock/KBOUND_WIN_BOOTSTRAP_CIS_oof.json`): no-harm safety results now backed by saved OOF confidence intervals rather than a pending replay. PACS, ImageNet-R, and CIFAR-10.1 are completed breadth tracks, not natural beats-both claims. The three-source mixture is a constructed routing aggregate; it is not a transfer result and does not change the per-dataset conclusion.

Theory and calibration. The conformal radius assumes exchangeable calibration pairs; sign bracketing without structure is impossible (Theorems 1 and 2; Appendix C). KGA's value scales with how often catastrophic, detectable harm occurs, not with universal accuracy gains. On helpful-dominated shifts the resulting near-*tie* with always-adapt reflects the certificate abstaining rather than committing harmful updates: KGA approximately matches the better policy, at most paying a small abstention cost when it declines a helpful but uncertifiable update, so the value proposition is conditional, like insurance against detectable harm. *Setting β.* The budget is *declared*, not estimated—a deployment-time upper bound on admissible calibration drift, fixed from domain knowledge, historical dev-to-deployment gaps, or an explicit stress envelope. The choice β=0 is the strongest zero-drift assumption and may be reported only as a plug-in baseline; it is not an assumption-free or conservative default. When no credible bound is justifiable, the population frontier cannot certify robustness to calibration drift, and the safe operational response is to abstain or freeze. Larger β widens the abstain band monotonically, trading decision coverage for robustness.

Detectability and transfer. ImageNet-R is a weak, split-specific evidence regime: one point-only arm does not survive both comparisons across three seeds. CIFAR-10.1 fails the cross-seed certificate bar ($\mathrm{FA_u} = 0.167$ and descriptive $\mathrm{FA_c} = 0.444$). These are useful boundary cases, not failed headline experiments: they show that a low-margin signal should not be promoted merely because one split or one candidate looks favorable.

Theory limits versus empirical limits. Some abstention is compatible with the population frontier, whereas baseline faithfulness, seed count, estimator quality, and replay status are ordinary empirical limitations requiring further validation. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding*: one cannot beat an oracle that is already always-adapt. Declining ImageNet-R is consistent with a weak-evidence, low-margin regime; the null result does not by itself establish structural non-identifiability under Theorem 2 (it could also reflect weak evidence, poor benefit estimation, or calibration-transfer failure). Abstention in the structurally ambiguous closed band prevents a uniformly sound strict commitment over the declared class. Empirical abstention, however, may instead stem from an overly wide radius or a weak estimator and therefore does not establish that a tested shift lies in that structural band. Richer label-free evidence may improve benefit estimation and reduce avoidable abstention, while better-justified or more data-efficient calibration can narrow finite-sample uncertainty; neither removes the population limitation without strengthening the declared assumptions or evidence.

Baseline scope and recomputation. On the ImageNet-C SAR configuration, the committal confidence gate adapts on harmful cells while KGA abstains on the low-margin band. The POEM/AETTA comparison is complete only for protocol-matched ports; official implementations remain future baseline work. The Office-Home/iWildCam summaries have a separate saved recomputation lock, but no external-lab replication; more seeds would strengthen ImageNet-C and PACS. Abstention lowers *decision* coverage but never withholds a prediction: it retains the frozen fallback f_0.

What we do not claim. Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; or wins on helpful-dominated shifts where always-adapt is already oracle-optimal.

XI. Reproducibility

A. Artifacts and Result Lineage

All locked/reconciled rows in Tables XV–XVI link to saved raw cells or seed aggregates in the anonymized repository. Provisional rows are retained only to document completed evidence and the required replay; they are not reused as headline claims. The core certificate module is packaged as an installable module in the anonymized repository.

B. Formal Verification Scope

The repository contains a kernel-checked Lean 4 formalization of indexed algebraic, finite-decision, measure-containment, and conditional uniform-rank results, with no `sorry`, `admit`, or project-defined axioms. It checks frontier sufficiency and the marginal false-adapt/false-freeze implication once coverage is assumed. The current development does not formalize the target-law witness construction, frontier necessity or maximality, risk alignment, multiclass identifiability, calibration transfer, or a general exchangeable-process lift. The implementation theorem uses the same exact rank rule as the current clean-split scorer; historical benchmark artifacts produced with an interpolated empirical quantile are not presented as kernel-certified split-conformal experiments. To our knowledge, machine-checked conformal-coverage results remain uncommon, but no priority claim is made.

C. Claim-to-Support Map

XII. Conclusion

K-Bound reframes test-time adaptation as a validity decision rather than an unconditional update. The central object is the target benefit $\Delta = R_T(f_0) - R_T(f_a)$. Over a declared drift-budget class, a strict adapt or freeze commitment is uniformly supportable if and only if the observable disagreement-region margin satisfies $|M| > \beta$; on $|M| \le \beta$, abstention is the maximal sound three-way action under the paper's strict-decision semantics.

KGA operationalizes this principle with a label-free benefit estimate and a pre-calibrated uncertainty radius. The resulting controller commits the candidate only when the interval excludes zero, retains the frozen model when harm is certified, and otherwise abstains from changing state. Across the benchmark panel, the observed behavior follows the regime map: controlled mixed regimes provide beats-both routing results, one-sided natural shifts primarily provide no-harm behavior, and weak or non-transferable evidence remains diagnostic rather than being promoted as a win.

The broader implication is that an adaptation system should expose two components: an update mechanism and a validity layer that decides whether the update is supportable under the available evidence and assumptions. Strengthening K-Bound therefore requires not only better adapters, but also richer risk-aligned evidence, more defensible calibration across shifts, explicit drift-budget auditing, and reproducible deployment semantics.

Appendix A
Calibration and held-out evaluation

Algorithm 2 gives the shared evaluation protocol behind the uniform nine-track panel (Table XV) and the primary numeric evidence (Table XVI).

Appendix B
Evidence feature schema and adapter hyperparameters

The promoted per-condition artifacts use the following label-free evidence panel. Notation: p_0, p_a are the frozen and adapted softmax rows on the adaptation batch; $\bar{m}_0 = \mathrm{mean}_n\, p_{0,n}$ and $\bar{m}_a$ are the predicted-class marginals; $H(p) = -\sum_c p_c \log p_c$; C is the number of classes. The exact 11-dimensional schema is implemented in `kbound_pkg/kbound/evidence.py`. The retained Camelyon17 multi-seed logs also record this same ordered 11-feature vector. Tracks available only through reconciled summary artifacts are not used to claim a richer feature schema.

Algorithm 2 Calibration and held-out evaluation for K-Bound

Require: Conditions $\mathcal{C}$; frozen f_0; adapters $\mathcal{A}_1, \ldots, \mathcal{A}_K$; calibration split $\mathcal{C}_{\mathrm{cal}}$; test split $\mathcal{C}_{\mathrm{test}}$; level α

Ensure: Regret-to-oracle, false-adapt rate, coverage, beats-both verdict

for all $c \in \mathcal{C}_{\mathrm{cal}}$ **do**
  for all adapters $\mathcal{A}_k$ **do**
    $f_a^{(k,c)} \leftarrow \mathcal{A}_k(f_0, X_c)$
    $Z_{k,c} \leftarrow \phi(X_c, f_0, f_a^{(k,c)})$
    $\Delta_{k,c} \leftarrow R_c(f_0) - R_c(f_a^{(k,c)})$
  end for
end for
Calibrate without test leakage: on stress grids, form leave-one-cell-out residuals $\rho_i = |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|$; on semantic shifts, reserve a development calibration split disjoint from the declared test split
$\varepsilon \leftarrow (1-\alpha)$ empirical quantile of the calibration residuals
Fit deployment estimator $\widehat{\Delta}(Z)$ on *all* calibration pairs ▷ scores the held-out test split
for all $c \in \mathcal{C}_{\mathrm{test}}$ **do**
  for all adapters $\mathcal{A}_k$ **do**
    Run Algorithm 1 with $(f_0, \mathcal{A}_k, X_c, \widehat{\Delta}, \varepsilon)$
    Record decision, regret-to-oracle, coverage, false-adapt indicator
  end for
end for
Aggregate metrics vs. always-adapt, always-freeze, per-condition oracle

---

Appendix C
Complete core proofs

This appendix is not required for the main empirical claim; it preserves the full theorem stack and validator mapping behind the three main-paper theorems: matched-evidence impossibility (Theorem 1), the benefit-sign frontier (Theorem 2), and the finite-sample certificate (Theorem 3). The main text contains the complete algebraic proofs; the material below records only the supporting matched-evidence form and zero-drift interpretation used by those proofs.

a) Core-theory inventory.: The main text gives complete proofs of disagreement reduction, interior impossibility, boundary abstention, the strict-commitment frontier, and the coverage-based certificate. This appendix records the matched-evidence full form and the zero-drift interpretation. Minimax, one-bit, capacity, and rate extensions are deferred and are not part of the short-paper claim stack.

Theorem 4 (Matched evidence impossibility (full form))**.** *Theorem 1 is part (iii) of the following. Fix $\alpha \in (0, \frac{1}{2})$.*

(i) *Witness worlds with* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ *and* $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$.

TABLE XXI

Claim-to-support map. "Theorem" means proved in the manuscript; Lean coverage is limited to the inventory described above. "Empirical" and "diagnostic" claims are restricted to the named saved artifacts and protocols.

Claim	Type	Exact requirement	Evidence location	Caveat
Interior matched-evidence impossibility	theorem	$\beta > 0$, $\lvert M\rvert < \beta$, declared class contains the constructed worlds	Thm. 1; App. C	Opposite nonzero benefits are an interior result only.
Closed-band abstention	theorem	Strict three-way soundness over $\lvert\gamma\rvert \le \beta$	Prop. 2	Boundary is zero-versus-strict ambiguity, not opposite signs.
Strict-commitment frontier	theorem	Declared drift class and risk alignment	Thm. 2	Uniform strict commitment, not universal sign recovery.
Marginal $\mathrm{FA_u}$ certificate	theorem	$\mathbb{P}(\lvert\widehat{\Delta} - \Delta\rvert \le \varepsilon) \ge 1 - \alpha$	Thm. 3; Lean inventory	Does not control conditional $\mathrm{FA_c}$.
CIFAR-10-C mixed-regime beats-both	empirical	Locked Tent/EATA stress aggregate and paired CIs	Table IX; canonical manifest	Current raw tree does not replay archived SAR; SAR withheld.
ImageNet-C authoritative result	empirical	27 cells, SAR candidate, paired condition bootstrap	Table XIII; bootstrap JSON	One seed; no cross-seed claim.
Natural no-harm results	empirical	Reconciled OOF or locked held-out artifacts	Tables XV–XVI	Not single-dataset natural beats-both.
POEM/AETTA comparison	empirical	Same logged stress cells and protocol-matched ports	Table XI	Not official implementations.
Constructed heterogeneous mixture	empirical	Researcher-constructed three-source routing aggregate	Sec. VII-I	Not unseen-shift transfer.
Universal improvement	diagnostic	No supporting requirement is declared	Sec. X	Explicitly not claimed.
Single-dataset natural beats-both	diagnostic	Would require held-out CI superiority to both fixed policies	Natural-shift tables	Explicitly not claimed.
Real-camera study	pending	Fresh preregistered held-out sessions and raw logs	pre-registration templates	Templates are not evidence.

TABLE XXII

Promoted 11-dim evidence panel.

Feature	Definition	Family
`pre_entropy`	$\mathrm{mean}_n H(p_{0,n})$	frozen output
`pre_conf`	$\mathrm{mean}_n \max_c p_{0,n}$	frozen output
`pre_pbal`	$H(\bar{m}_0)/\log C$	frozen marginal
`post_entropy`	$\mathrm{mean}_n H(p_{a,n})$	adapted output
`post_conf`	$\mathrm{mean}_n \max_c p_{a,n}$	adapted output
`post_pbal`	$H(\bar{m}_a)/\log C$	adapted marginal
`pbal_drop`	`pre_pbal`$-$`post_pbal`	change
`entropy_drop`	`pre_entropy`$-$`post_entropy`	change
`frac_highconf`	$\mathrm{frac}_n(\max_c p_{a,n}{>}0.9)$	collapse
`marginal_KL`	$\sum_c \bar{m}_{a,c}\log(\bar{m}_{a,c}/\bar{m}_{0,c})$	drift
`update_norm`	$\lVert\Delta\theta_{\mathrm{aff}}\rVert_2$	update

TABLE XXIII

Per-track adapter hyperparameters (from the run configs). Adapted parameters are BN/LayerNorm-affine only; TTA optimizer is Adam. "Aggressiveness" cells use mild = (10 steps, lr 10^{-3}) or aggressive = (50 steps, lr $2.5{\times}10^{-3}$). EATA uses an entropy filter $e_{\mathrm{thr}}{=}0.4\ln 2{\approx}0.28$; SAR adds SAM + entropy reset. Backbones for CIFAR-10-C, ImageNet-C, PACS, and Camelyon17 are verified from the run code; the remaining WILDS/DomainBed tracks use the standard baseline.

Track	Backbone	Adapter(s)	Steps / lr
CIFAR-10-C stress	ResNet-18	Tent/EATA/SAR	mild & aggressive (per cell)
ImageNet-C	ResNet-50	Tent/EATA/SAR (SAM+reset)	mild & aggressive (per cell)
PACS (LODO)	ResNet-18	mild/aggressive	$10/10^{-3}$ & $50/2.5{\times}10^{-3}$
Camelyon17	DenseNet-121	EATA online	10 / 10^{-3}
iWildCam	ResNet-50	Tent episodic	10 / 10^{-3}
Office-Home	ResNet (DomainBed)	SAR online aggr.	50 / $2.5{\times}10^{-3}$
RxRx1	ResNet-50	episodic	5 / (batch 64)

(ii) *Le Cam identity:* $\inf_g M(g) = 1 - \mathrm{TV}$ *between the evidence laws.*

(iii) *Forced abstention at rate* $\ge 1 - 2\alpha$ *under dual error control.*

Corollary 2 (Closed-band abstention under dual error control)**.** *In the* $\lvert M\rvert \le \beta$ *wedge of Theorem 2, Theorem 4(iii) implies any label-free rule with false-adapt and false-freeze* $\le \alpha$ *must abstain with probability at least* $1 - 2\alpha$ *on matched evidence. Opposite nonzero signs are required only for* $\lvert M\rvert < \beta$*; at* $\lvert M\rvert = \beta > 0$ *zero-versus-strict ambiguity suffices, and at* $M = \beta = 0$ *the benefit is identically zero so neither strict action is sound.*

Remark 3 ($\beta = 0$ face)*.* The $\beta = 0$ face assumes $\gamma = 0$. Under the stated disagreement-score decomposition, strict commitment is sound for $M \ne 0$, while $M = 0$ yields $\Delta = 0$. Existing estimators may be interpreted as $\beta = 0$ plug-ins only when their estimand realizes this decomposition; $\gamma = 0$ alone does not establish their soundness.

Appendix D
Multiclass and regression derivations

For multiclass 0/1 loss, errors cancel outside $D = \{f_0 \ne f_a\}$, giving $\Delta = \mu_T(D)(p_a - p_0)$ with $p_j = \mathbb{P}_T(f_j = Y \mid D)$. In contrast to binary classification, p_0 need not equal $1 - p_a$. If $p_a - p_0 = M_{\mathrm{mc}} + \gamma_{\mathrm{mc}}$ and $\lvert\gamma_{\mathrm{mc}}\rvert \le \beta_{\mathrm{mc}}$, then $\lvert M_{\mathrm{mc}}\rvert > \beta_{\mathrm{mc}}$ is sufficient for a strict commitment; necessity additionally needs the declared class to realize the admissible evidence-identical drifts.

For squared-loss regression, let $m_T(x) = \mathbb{E}_T[Y \mid X = x]$. Direct expansion gives

$$\Delta = \mathbb{E}_T[(f_a - f_0)(2m_T - f_0 - f_a)]\,.$$

The integrand vanishes where $f_a = f_0$, so the comparison is again disagreement-local. For an observable proxy $u(x)$, define

$$M_{\text{reg}} = \mathbb{E}_T[(f_a - f_0)(2u - f_0 - f_a)],$$
$$\gamma_{\text{reg}} = 2\mathbb{E}_T[(f_a - f_0)(m_T - u)].$$

Then $\Delta = M_{\text{reg}} + \gamma_{\text{reg}}$. A declared bound $|\gamma_{\text{reg}}| \leq \beta_{\text{reg}}$ yields the same sufficient strict-commitment rule $|M_{\text{reg}}| > \beta_{\text{reg}}$; a converse again requires a richness assumption on the target class.

Appendix E
Historical protocol reconciliation

The main text contains the complete empirical accounting in the uniform nine-track panel. Historical route studies and superseded per-dataset tables are deliberately excluded from this short-paper build. Their raw logs remain in the repository for audit, but they are not independent empirical claims. The declared evaluation unit is a condition or held-out cell, never an arbitrary pool of frames or domains; all policies replay the same unit stream, and labels are revealed only after the KGA decision for offline scoring.

Appendix F
Physical-study pre-registration

The two-phone package-inspection protocol is locked before fresh capture. Table XXIV records the split and deployment design; Table XXV fixes the primary endpoint layout. Every result cell remains blank until the exact physical inventory, source-model gate, development seal, held-out chronology, replication replay, and anti-leakage audit pass the machine-readable publication gate. Browser previews, mock clips, and reconstructed media are software diagnostics, not evidence.

Appendix G
Runtime details

The adapter update requires forward and backward computation before the controller can decide. The saved controller microbenchmark reports evidence extraction, benefit-model evaluation, rollback copy memory, and benefit-model size; these are the added-cost quantities reported in Table VI. A complete end-to-end profile separating frozen inference, candidate adaptation, candidate inference, rollback time, and total window latency remains pending and is not inferred from the controller-only microbenchmark.

Appendix H
Formalization inventory

Lean checks the indexed algebraic, finite-decision, measure-containment, and conditional uniform-rank results listed in `formal/KBound/TheoremMap.lean`. It does not formalize the target-law witness construction, frontier necessity/maximality, risk alignment, calibration transfer, or the lift from a conditional uniform-index model to general exchangeable processes.

Appendix I
Claim-to-artifact map

The machine-readable manifest `paper/generated/kbound_result_manifest.json` is the authoritative index for every promoted number, seed count, protocol, quantile convention, and known replay caveat. Table XXI gives the reader-facing summary; the external manifest retains exact repository paths and pending work.

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TABLE XXIV

R1 – Pre-registered real-world deployment protocol. The split and online-label constraints are fixed before physical held-out capture. Session identifiers are populated from the signed recording manifest.

Item	Locked design
Task	Four-class package/label inspection: correct, missing label, misaligned label, damaged label
Camera	Two phone cameras and real physical objects; the second phone is reserved for external-device replication
Base model	MobileNetV3-Small
Candidate adaptation	Episodic Tent; one update step per decision window
Decision window	32 frames
Evidence	14 label-free frozen/candidate disagreement and shift features
Calibration-fit sessions	— (distinct recorded session; IDs locked in manifest)
Conformal calibration sessions	— (different day/session from calibration-fit)
Held-out physical test sessions	— (different day/session; inaccessible during tuning)
Physical shifts	Lighting, shadow, blur, background, viewpoint, distance, and batch composition
Policies compared	Always-freeze, always-adapt, confidence gate, entropy gate, KGA without radius, KGA certificate
Deployment labels used live?	No; labels are joined only during offline evaluation
Miscoverage level	$\alpha = 0.10$
Official live model	Frozen fallback unless the candidate is certified for adaptation

TABLE XXV

R2 – Primary held-out real-world result (RESULT PENDING). All policies will be scored on the identical locked physical stream after real captures pass the source-model gate (≥ 0.80 balanced accuracy and macro-F1 on S02). Until then every metric cell reads *RESULT PENDING — NO MEASURED DATA AVAILABLE*; development replays on mock clips are audit-only and must not be cited.

Policy	Balanced acc. ↑	Macro-F1 ↑	Regret ↓	FA_u ↓	FA_c ↓	Adapt rate	Abstain rate	Mean latency (ms) ↓
Always-freeze	—	—	—	—	—	—	—	—
Always-adapt	—	—	—	—	—	—	—	—
Confidence gate	—	—	—	—	—	—	—	—
Entropy gate	—	—	—	—	—	—	—	—
KGA, no radius	—	—	—	—	—	—	—	—
KGA certificate	—	—	—	—	—	—	—	—

$\text{FA}_u = \mathbb{P}(\text{adapt} \wedge \Delta \leq 0)$ is the certificate-aligned unconditional false-adapt rate; $\text{FA}_c = \mathbb{P}(\Delta \leq 0 \mid \text{adapt})$ is descriptive. Ground-truth labels are unavailable during live decisions and are revealed only for offline evaluation. Latency reporting includes candidate adaptation, the KGA decision, and full end-to-end window latency.

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